

Ramanath Amith

| Bengaluru, IND | ramanathamith@gmail.com | +91 9448201595 | [GitHub](#) |

Experience

Data Intern (SAR Data Processing) - GalaxEye Space (Sep 2023 - Dec 2023)

(Skills: Python | Digital Signal Processing | Synthetic Aperture Radar)

- Built a simulator to produce raw SAR (Synthetic Aperture Radar) images and performed Range and Azimuth compression. Experimented, analyzed Back projection algorithm.
- Implemented and tested an image enhancement algorithm for SAR images which maps a slant range SAR image to a ground range SAR image (Orthorectification, using SRTM 30m DEM).

Associate Software Engineer - Mphasis (Feb 2021 - Aug 2022)

(Skills: Java, Angular, Spring Boot, SQL, Shell scripting, Git, Jenkins, Docker)

- Worked on migration of a financial risk management system from SAS(Monolithic architecture) to a microservice based architecture.
- This system had several financial auditing algorithms (batch jobs) that accessed datasets/tables from various business groups across the organization. These jobs have a particular runtime schedule at which they are initiated (Ex: Job A runs on Mon, Thu, Sat at 7:15AM, Job B runs on Fri, Sat, Sun at 2:00AM). And also, these batch jobs have certain operations that work on dataset/tables which are common(Job A and Job B can depend on the same Table T1, Table T2).
- Some of the challenges I faced are: (1) Understanding the redundancy of operations on datasets in different batch jobs, (2) Understanding the runtime schedule of each job and if they are different, that require storing the intermediate results on Amazon S3 bucket (3) Building a dashboard that can monitor the status of these jobs and set the runtime schedule
- As a result, this reduced the usage of computational resources by 20%. And built a dashboard for monitoring the batch jobs(status, time settings, output settings).
- Wrote the boilerplate code for the project(defined the code structure, and wrote utility library)
- Drafted software design document and other reference documents (software architecture, code structure and deployment procedure).

Freelance IoT Developer (Sep 2020 - Dec 2020)

- Built a system to remotely monitor home/office which consists of an IP camera interfaced with Raspberry PI that is connected to a web server. When a person comes near the door, the system detects this using a motion sensor and starts recording a short video. This video is uploaded to the server and a notification is sent to the user.

Projects

Customer Relationship Management Application:

(Skills: Spring Boot, Angular, MySQL)

- A CRM tool that helps you manage customer queries/issues, assign tickets to support agents, analyze and monitor performance through an analytics dashboard.
- Admin can track tickets, agents performance (in terms of tickets assigned, response time, resolution time, SLA% and other parameters) and track clients (person, company, CSAT score, no. of tickets raised)
- Agents can get tickets assigned by Admin, respond to Clients and they can re-assign to other agents

- Clients can create and monitor tickets which are like interaction with agents.

Musician Bot: ([Code](#) | [Test Run Video](#))

(Skills: *Python, DSP(Audio), Embedded C*)

- A robot that understands music and plays it on an instrument.
- A file containing a series of musical notes played with 4 instruments, separated with short duration of silence, is given as input. The robot processes the file and extracts(Onset of a note, what note it is and which instrument it is played on.) and with this information, the bot has a mechanism which plays it on the actual instrument.
- There are three aspects to this project: Audio Processing(Onset, Instrument detection and note detection), Bot structure(physical structure), Controls and communication. I was responsible for Audio Processing part
- This was part of a competition ERYC-2018(conducted by ERTS Labs, IIT Bombay) . We were ranked 4th among 150+ teams.

Biped Patrol Bot: ([Code](#) | [Test Run Video](#))

(Skills: *Control Systems Design, Arduino, Embedded C*)

- A self-balancing robot that is remote controlled designed to move in uneven terrains and slopes. The design concept used to model the robot is an LQR Controller for an inverted pendulum, and for implementation we used Arduino Mega, MPU6050/GY-80, Xbee TX/RX. This project was part of a robotics competition ERYC-2019 (conducted by ERTS Labs, IIT Bombay) as a result we were in the top 10 teams among 150+ teams.

Languages

- English (Working proficiency)
- Hindi (Working proficiency)
- Tamil (Intermediate)
- Kannada (Intermediate)

Education

- Bachelor's In Engineering (*B.E. in Electronics and Communication Engineering*)
Nitte Meenakshi Institute of Technology - (Aug 2016 - Aug 2020, Score: 7.3CGPA):
- Class 12: State board (78%)
- Class 10: ICSE (80%)